measuring and experimenting

hands-on computer science · prof. dr. nicolas meseth

1. measuring settles it
2. know what you measure
3. one reading is a guess
4. build evidence

part 1

measuring settles it

which one lands first?

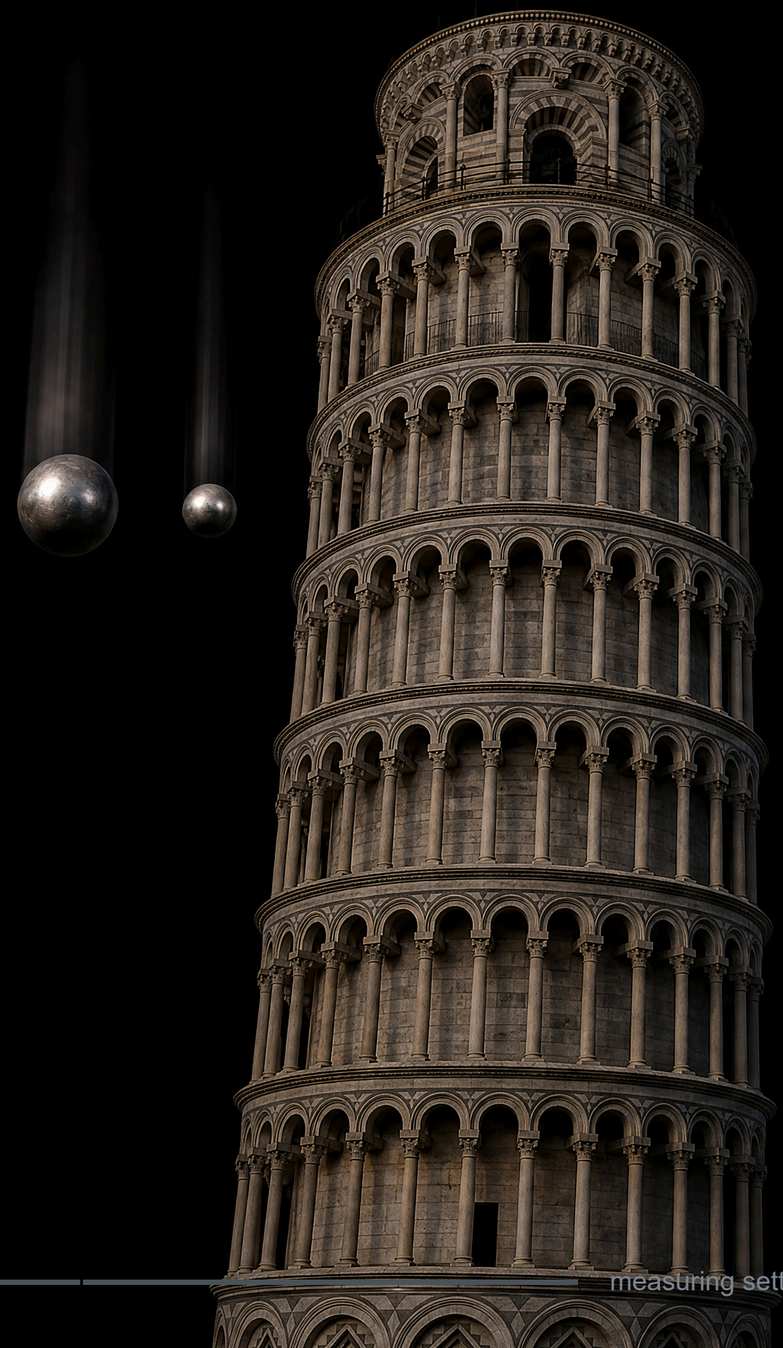


image: ai-generated (gpt-image-2)

three ways to answer the same question

it sounds right

longer readings
collect more light

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collect more light

the assistant predicts it

```
double the
integration time
```

three ways to answer the same question

it sounds right

longer readings
collect more light

the assistant predicts it

`double the integration time`

the series shows it

`41 / 50 → 47 / 50`
`recognised correctly`



evidence from your setup

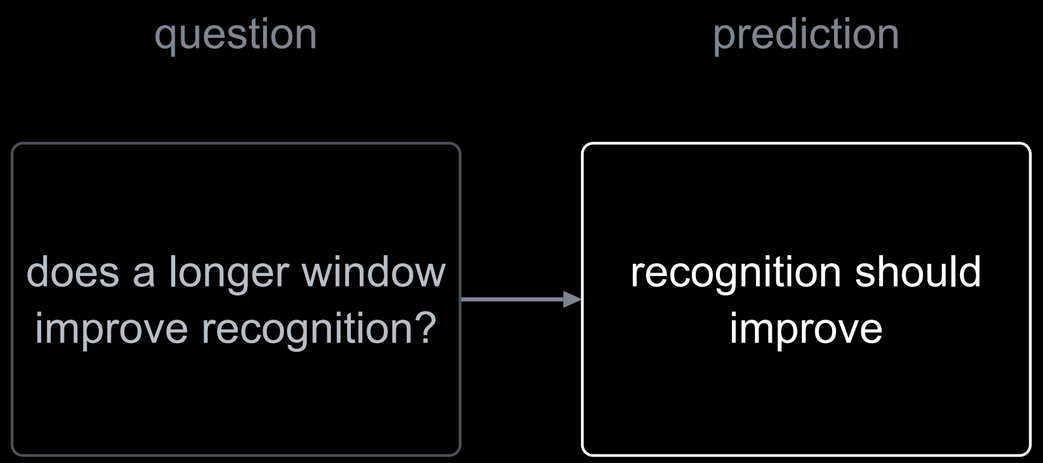
an explanation only claims something, while a measurement can decide it.

turn a claim into a test

question

does a longer window improve recognition?

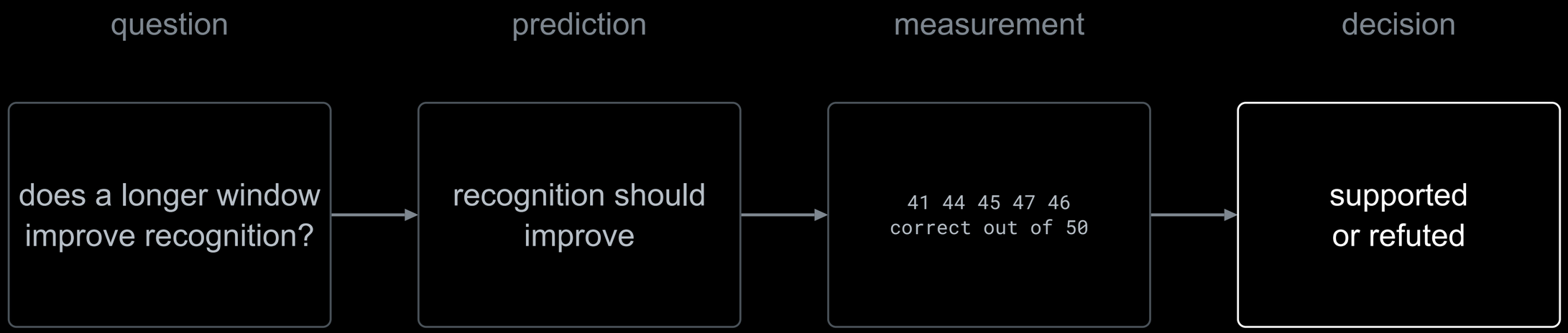
turn a claim into a test



turn a claim into a test



turn a claim into a test



write the expectation before you measure.

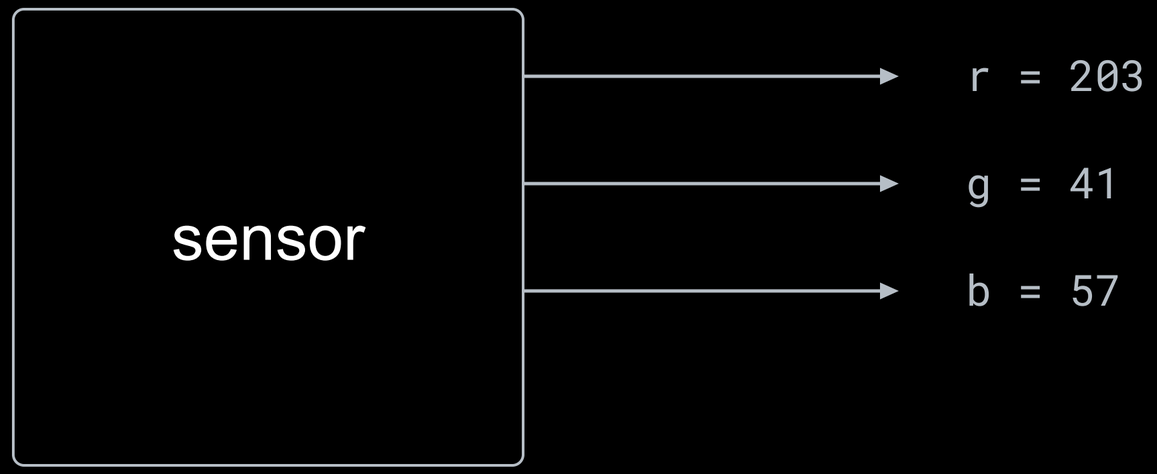
part 2

know what you measure

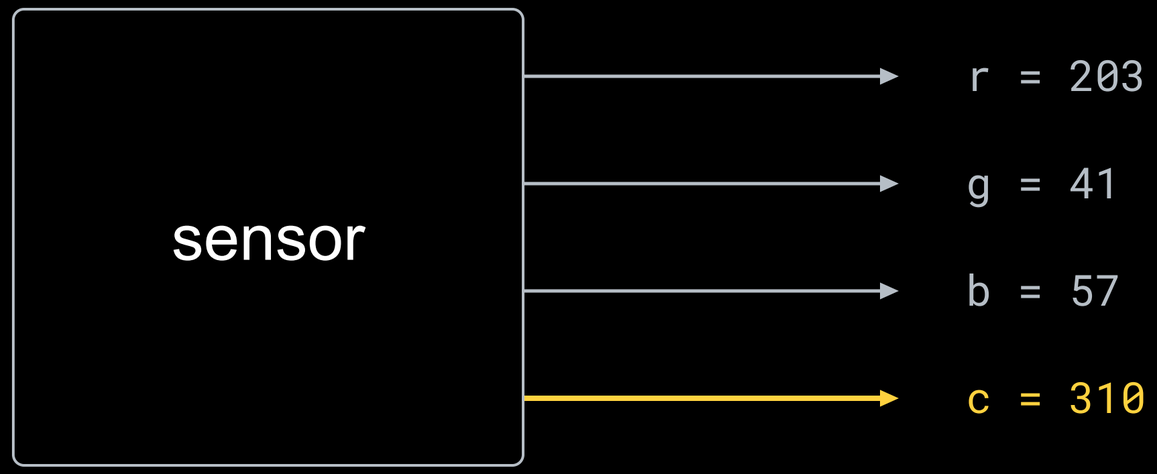
this concept lives at the input



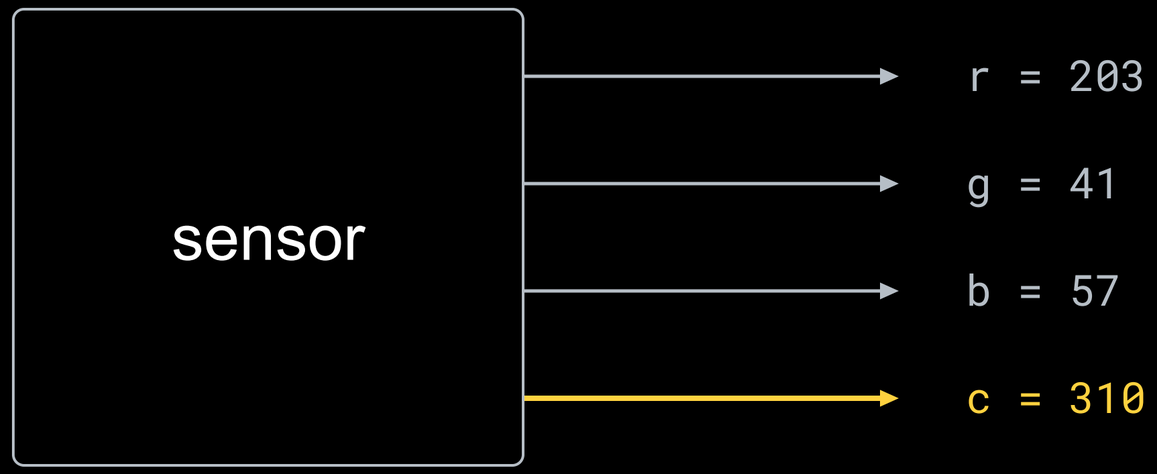
the sensor returns four numbers



the sensor returns four numbers



the sensor returns four numbers



`clear`

no colour filter

most sensitive

your eyes see a colour, but the program only sees four numbers.

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```


what arrives

```
r = 203
g =  41
b =  57
c = 310
```

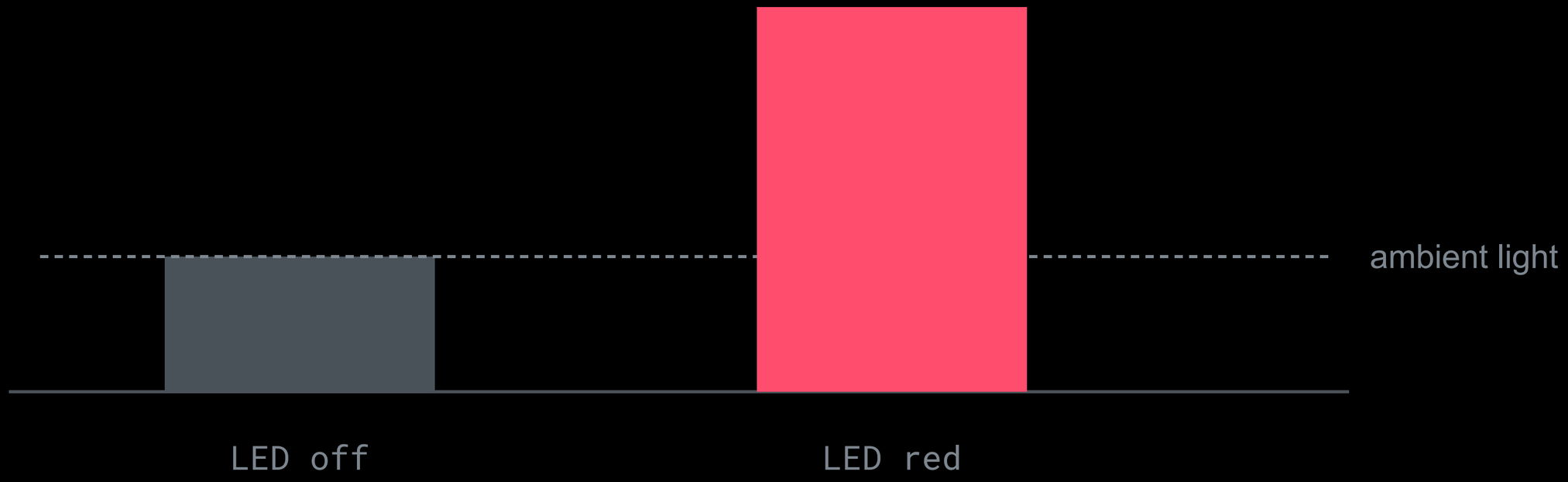
the filters overlap, distance dims the light, and the room shines in too

the numbers differ from what was sent, and that is normal, not a defect.

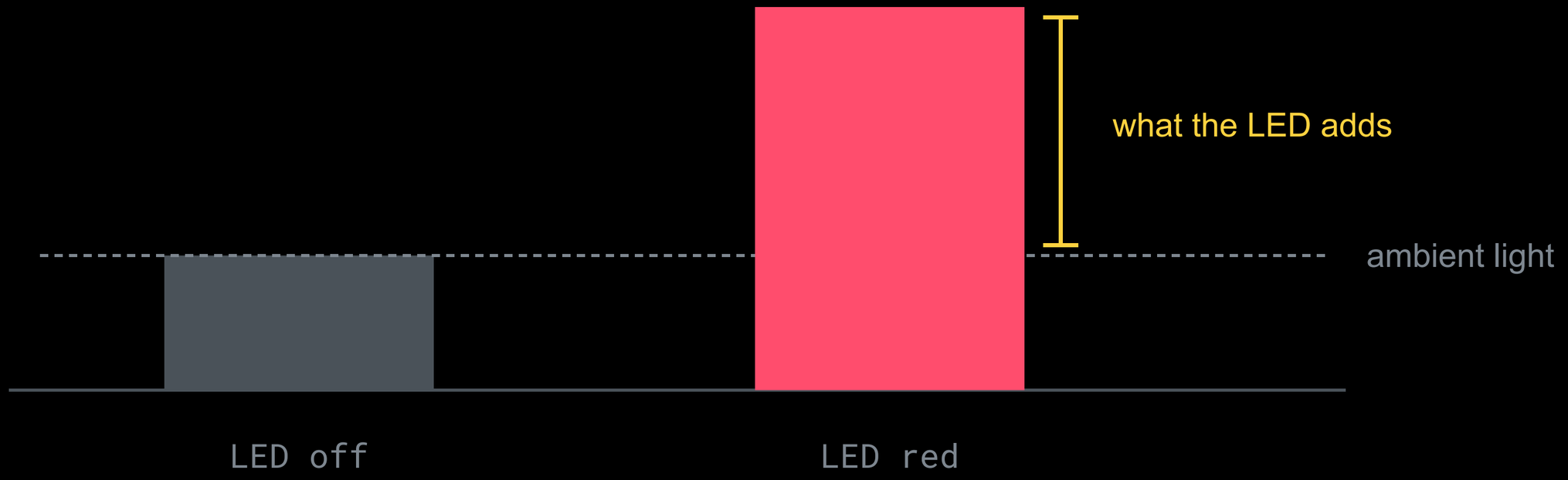
measure the dark first



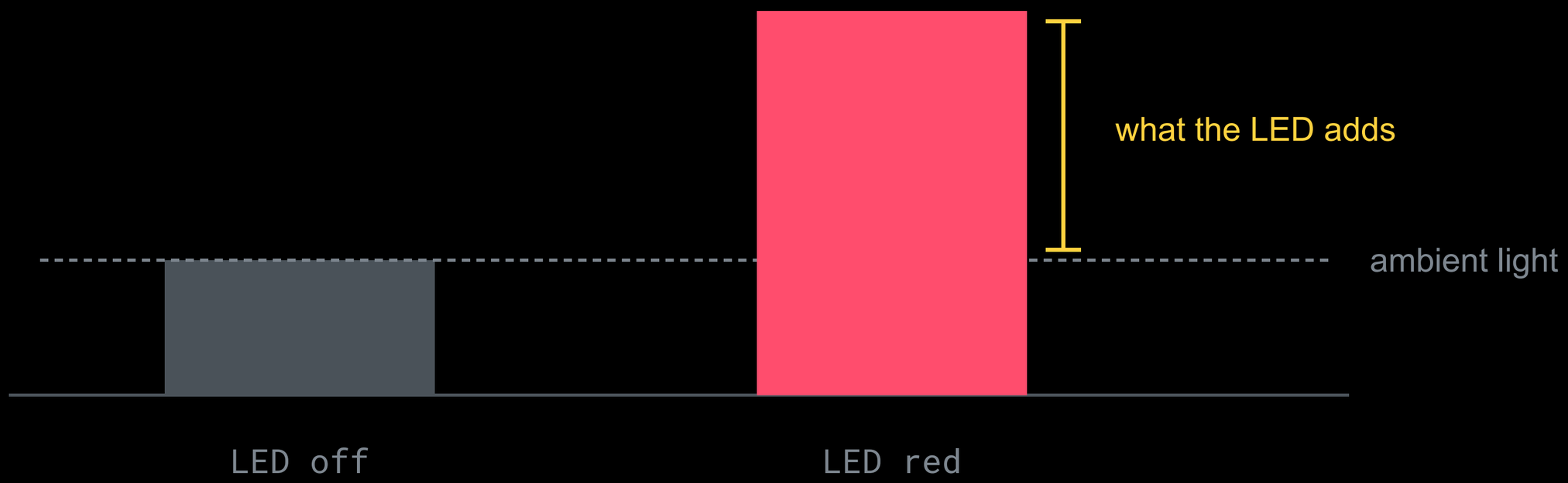
measure the dark first



measure the dark first



measure the dark first




```
reading = ambient + LED + noise
```

part 3

one reading is a guess

which value is true?

198

which value is true?

198

205

one reading is a guess

which value is true?

198 205 201 195 206

which value is true?

198 205 201 195 206

same LED · same settings · five readings

one reading is a guess

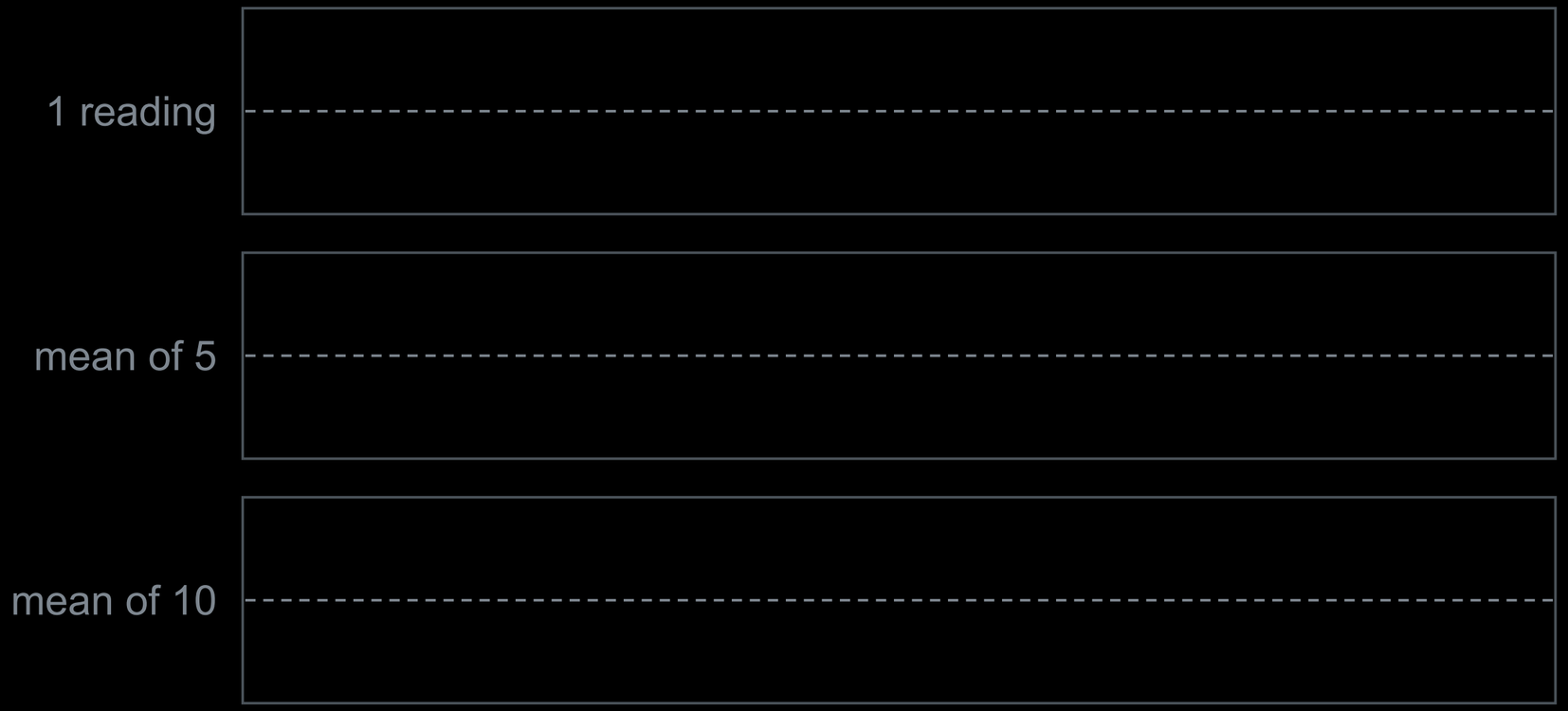
which value is true?

198 205 201 195 206

same LED · same settings · five readings

all of them are measurements

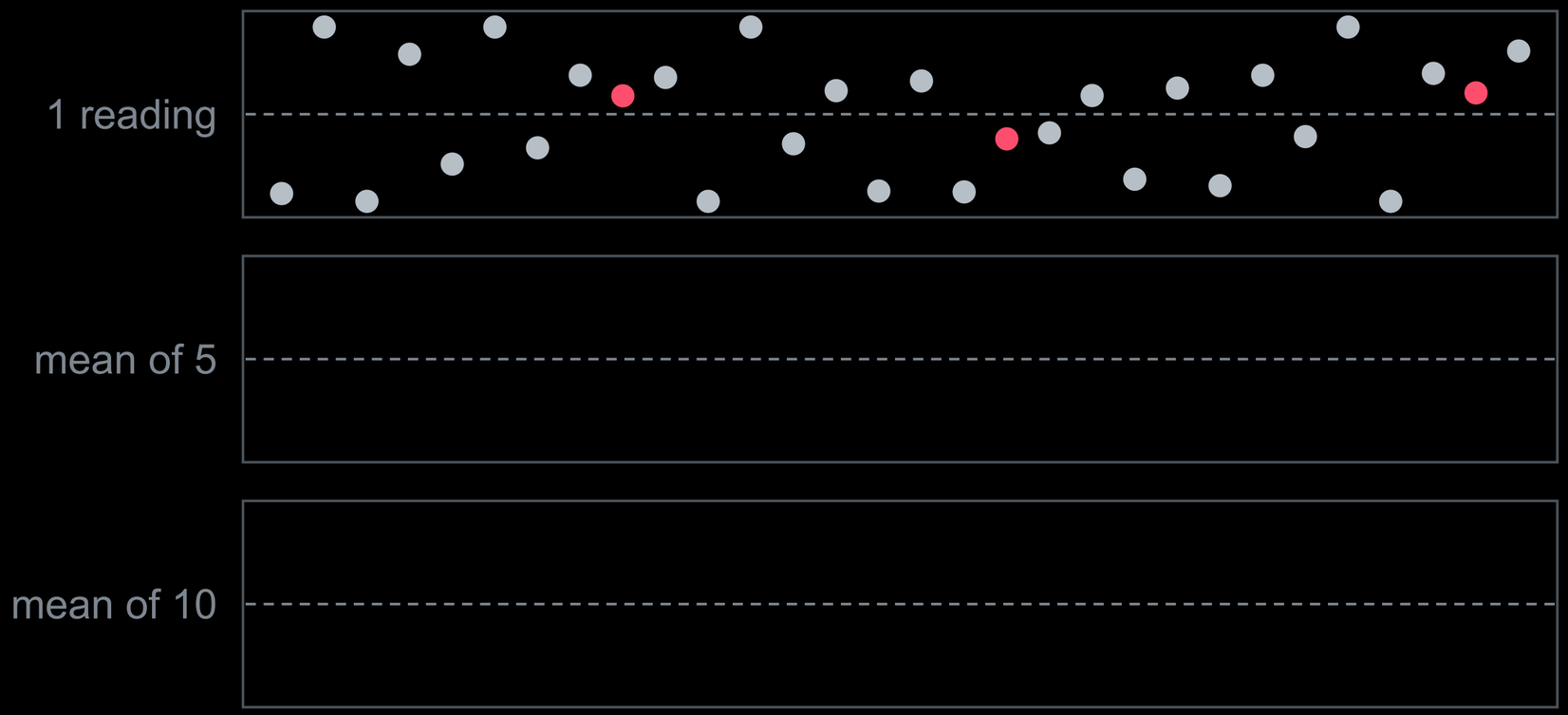
how many readings per decision?



a dot on the wrong side of the dashed threshold is a wrong decision

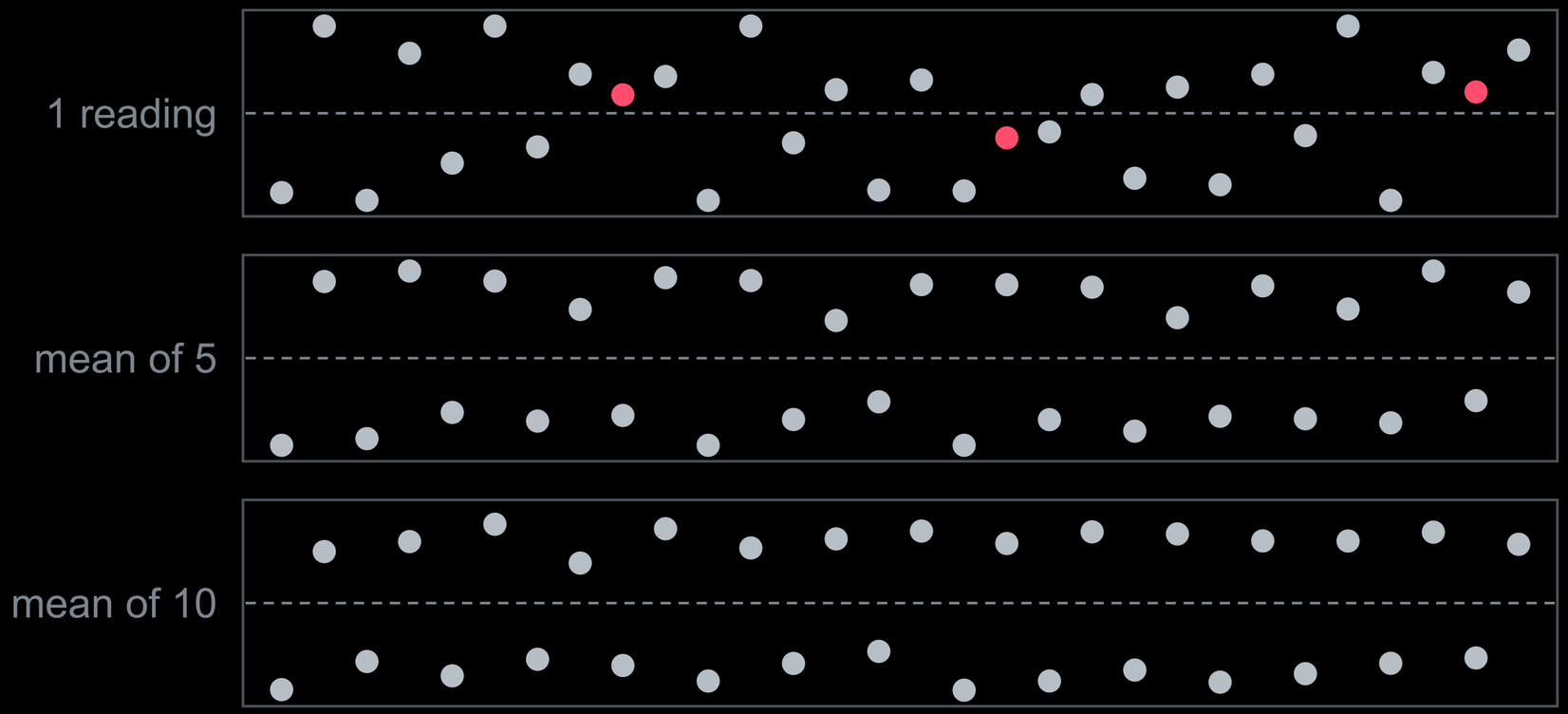
one reading is a guess

how many readings per decision?



a dot on the wrong side of the dashed threshold is a wrong decision

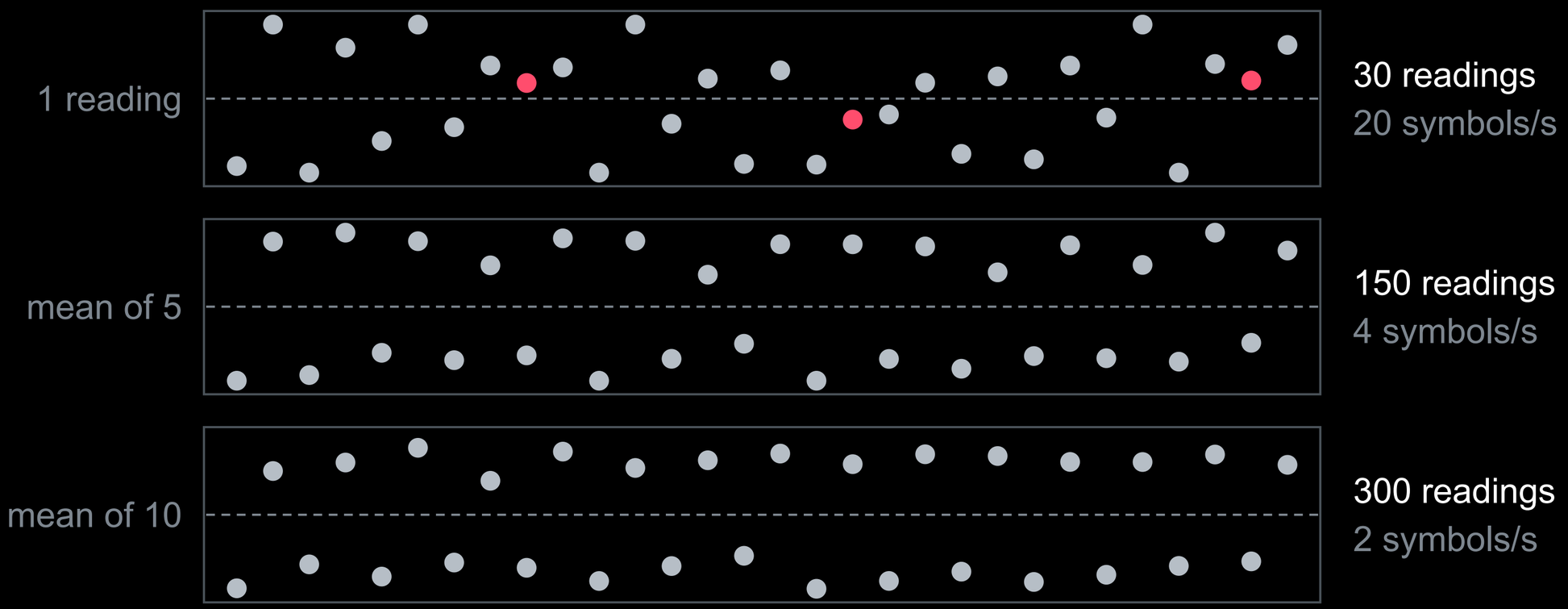
how many readings per decision?



a dot on the wrong side of the dashed threshold is a wrong decision

one reading is a guess

how many readings per decision?

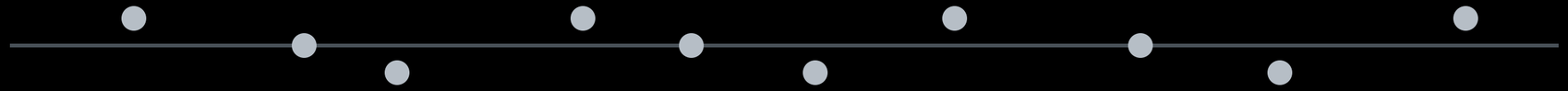


a dot on the wrong side of the dashed threshold is a wrong decision

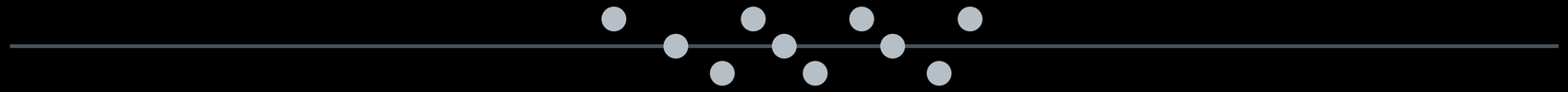
averaging lowers the error rate, but you pay for it with speed.

a series reveals what one reading hides

individual readings



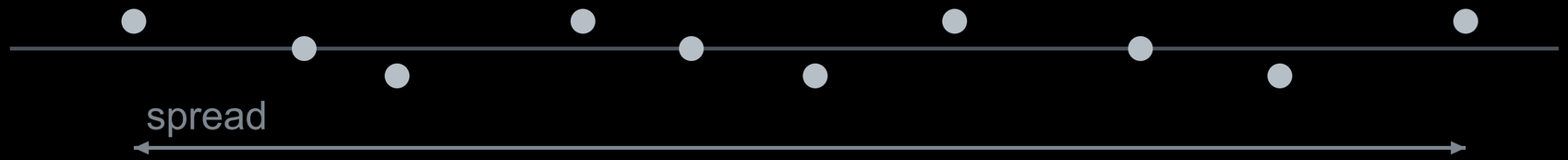
averages of five



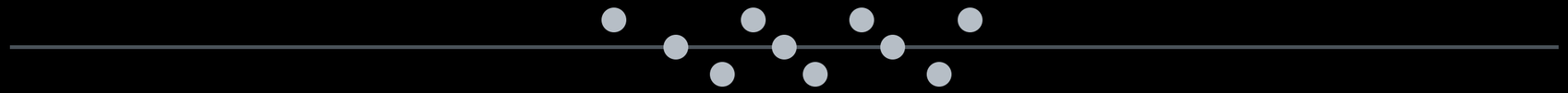
one reading is a guess

a series reveals what one reading hides

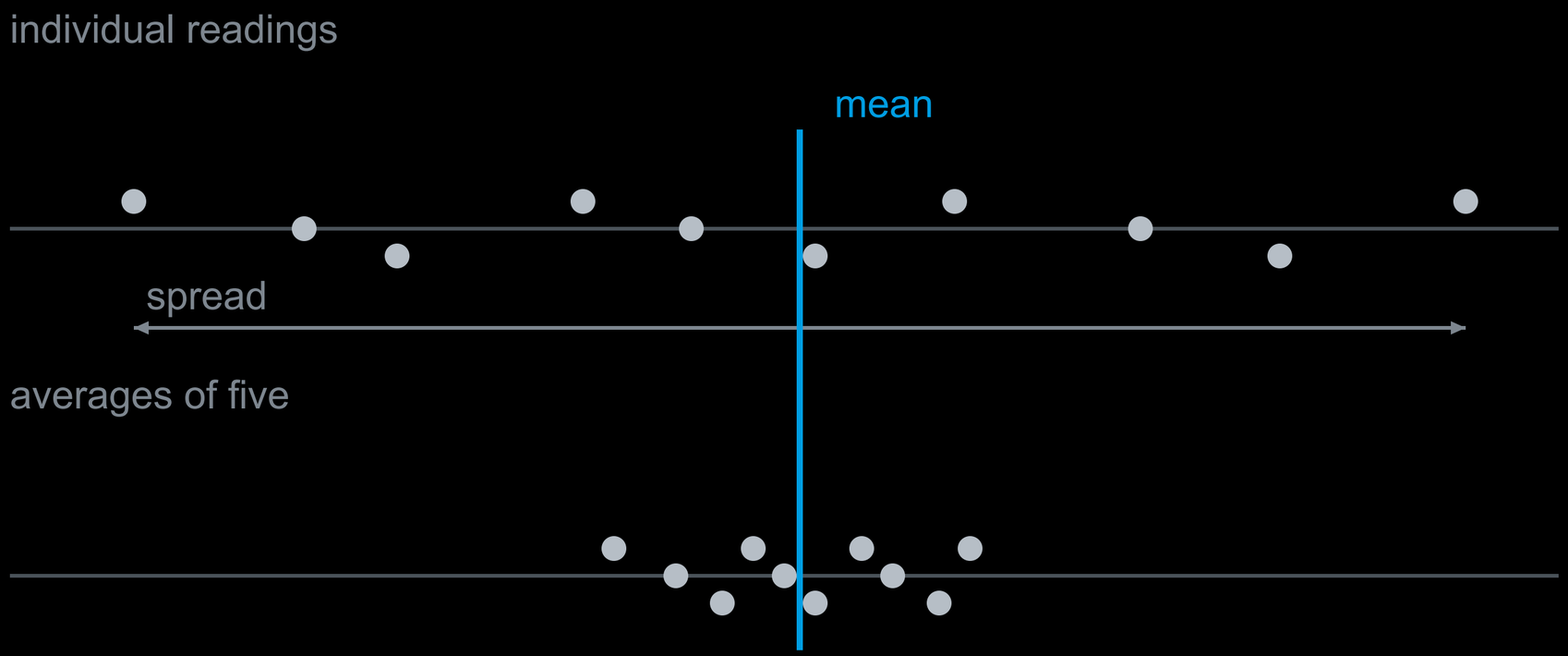
individual readings



averages of five



a series reveals what one reading hides



report the centre and the spread.

the knob: integration time



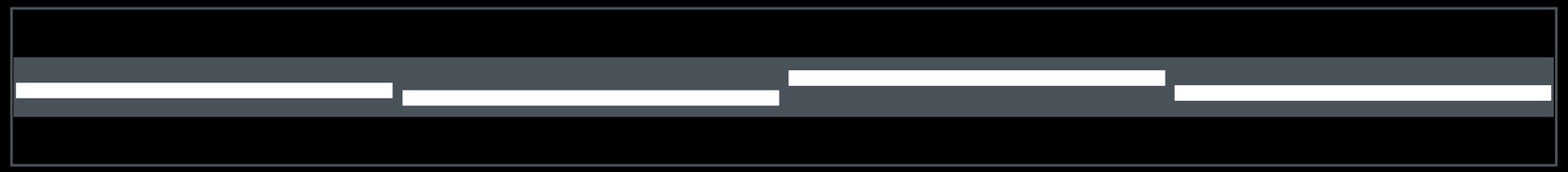
one reading is a guess

the knob: integration time

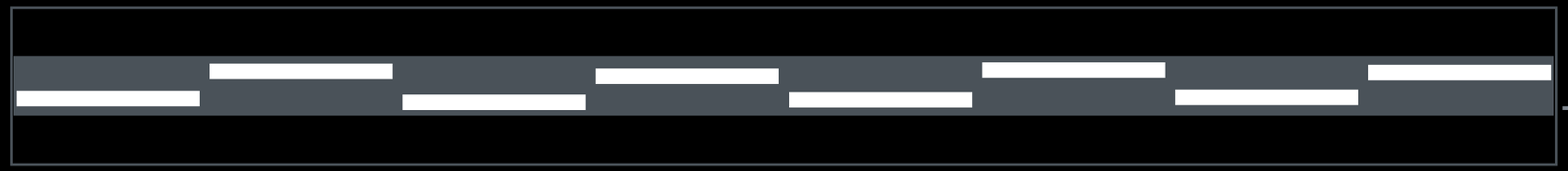


the knob: integration time

few long readings



more, shorter readings



the band that still reads correctly

many short readings



these land in the wrong colour

one second

one reading is a guess

a longer window gives calmer readings, a shorter one gives them faster.

where the best trade-off lies for your setup, only a measurement can tell.

part 4

build evidence

two teams improve their result

team a

changed distance and integration time

recognition improved

team b

changed only distance

recognition improved

two teams improve their result

team a

changed distance and integration time

recognition improved



what caused it?

team b

changed only distance

recognition improved



`distance`

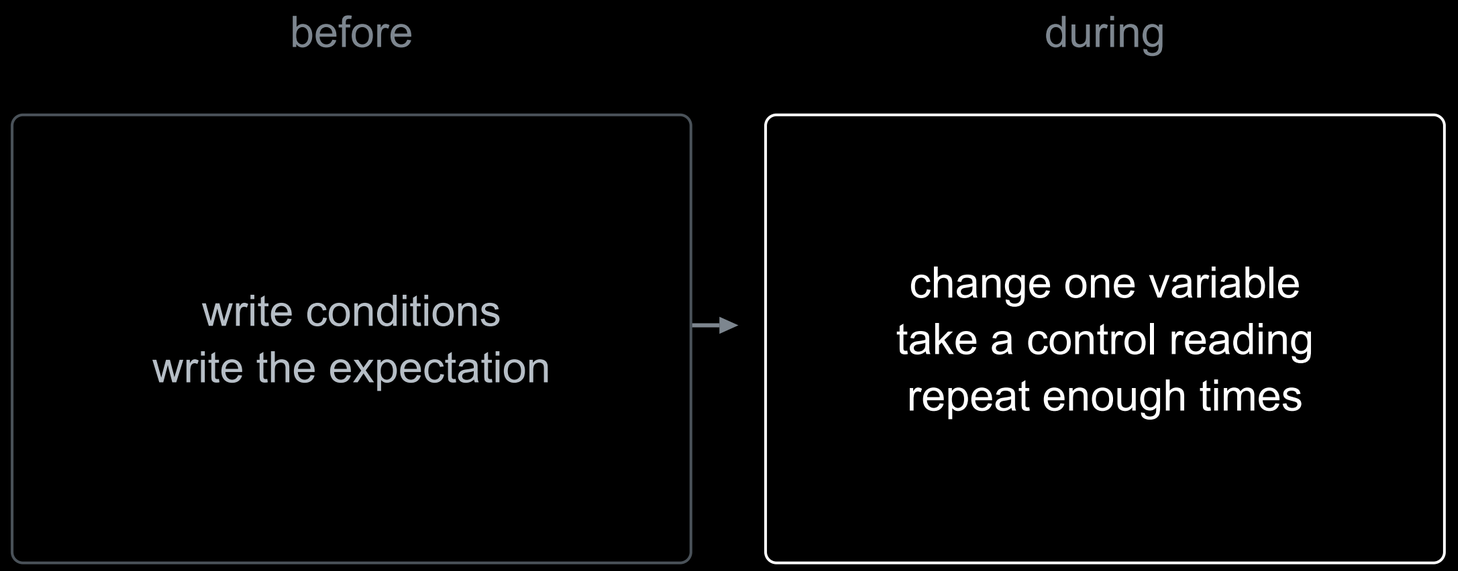
change one variable at a time.

what makes a measurement series

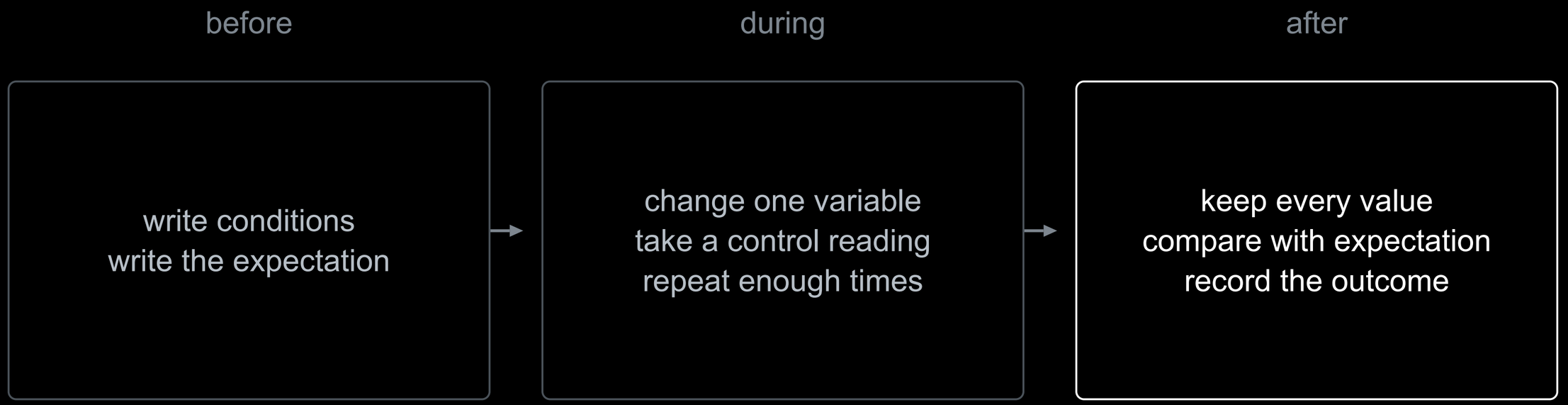
before

write conditions
write the expectation

what makes a measurement series



what makes a measurement series



a repeatable result is worth more than an impressive one.

luck or skill?

5 / 5

50 / 50

guessing succeeds about once in 32 attempts

luck or skill?



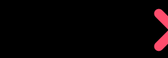
guessing succeeds about once in 32 attempts



guessing: about 1 in 1,125,899,906,842,624

luck or skill?

5 / 5



guessing succeeds about once in 32 attempts

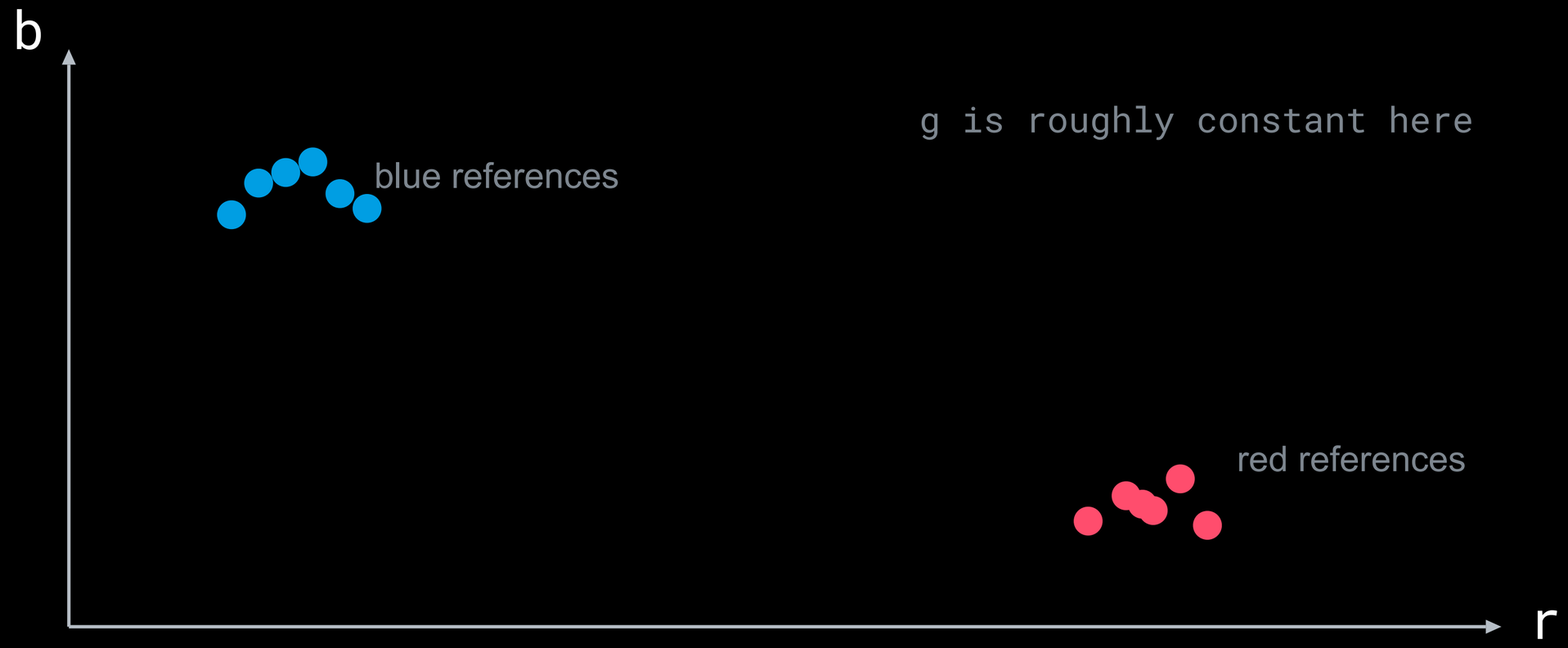
50 / 50



`guessing: about 1 in 1,125,899,906,842,624`

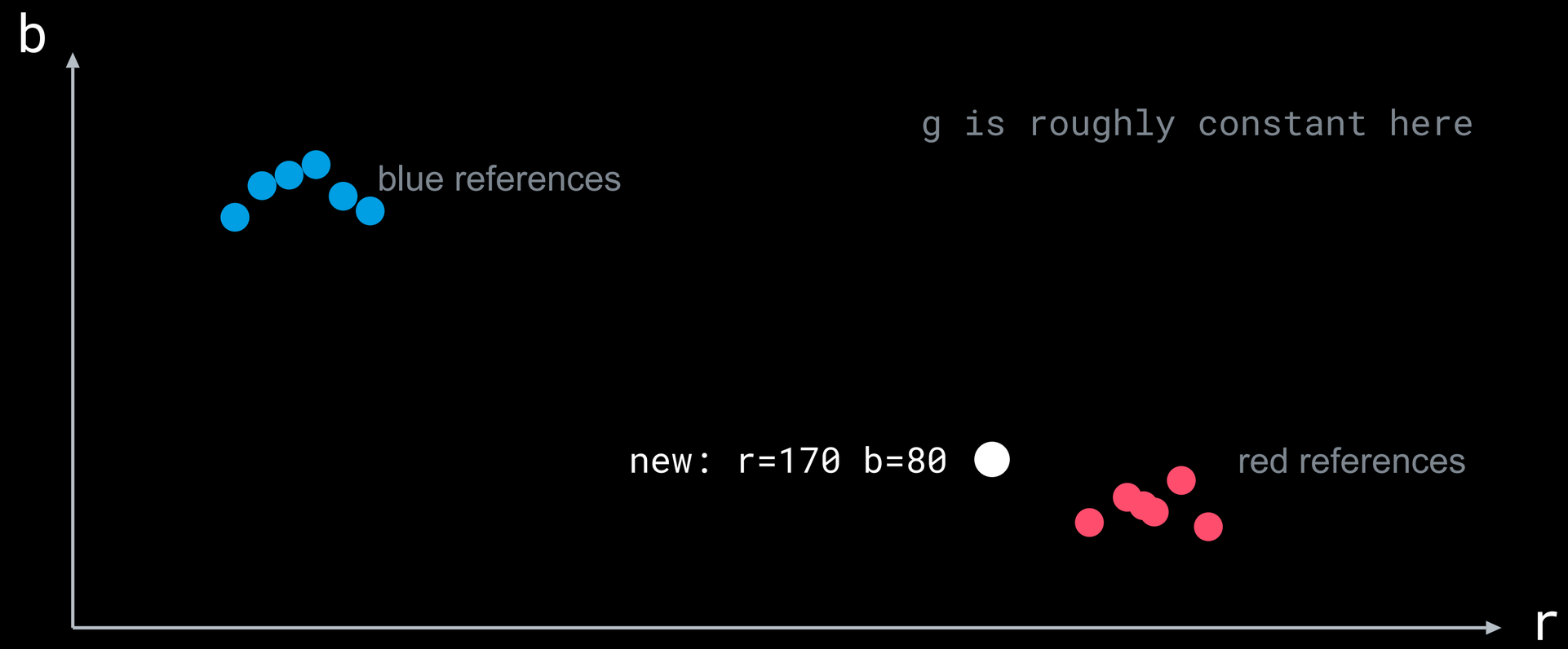
evidence

calibrate, then classify



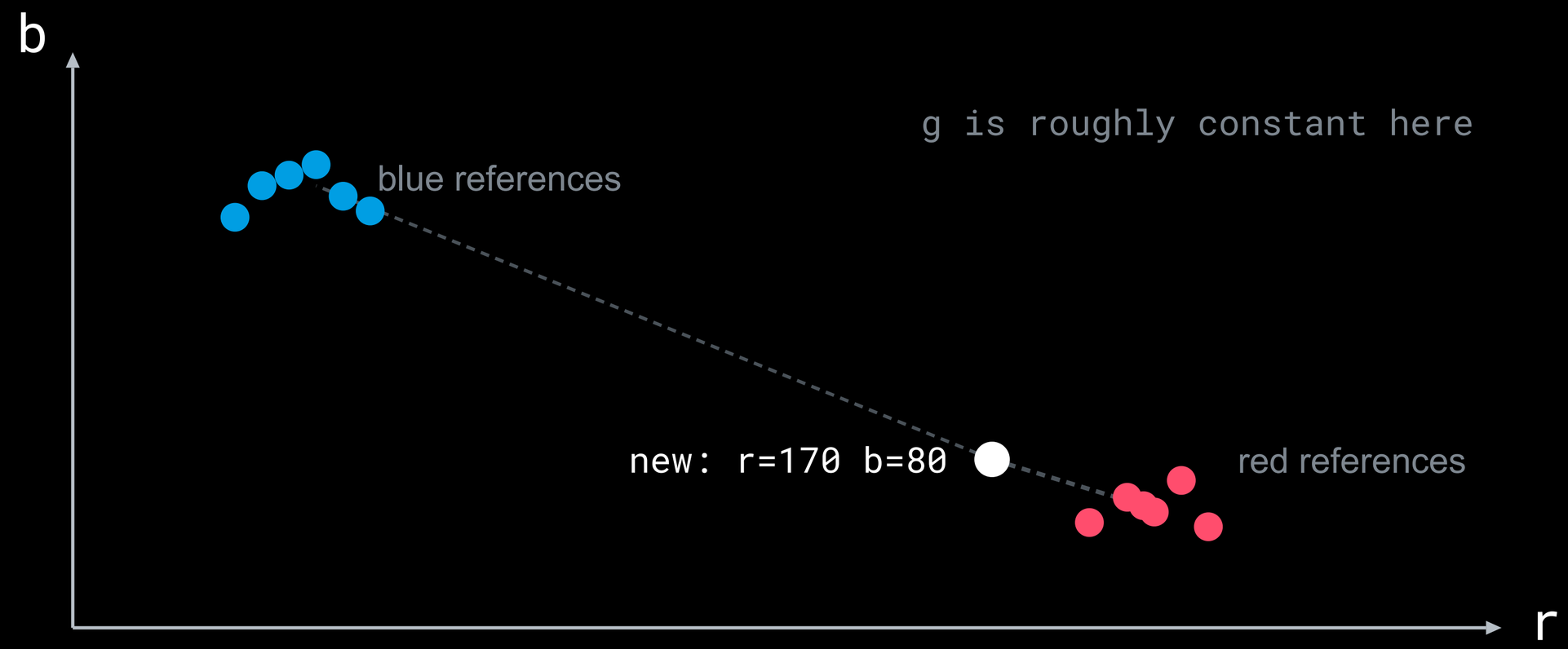
measure known colours many times

calibrate, then classify



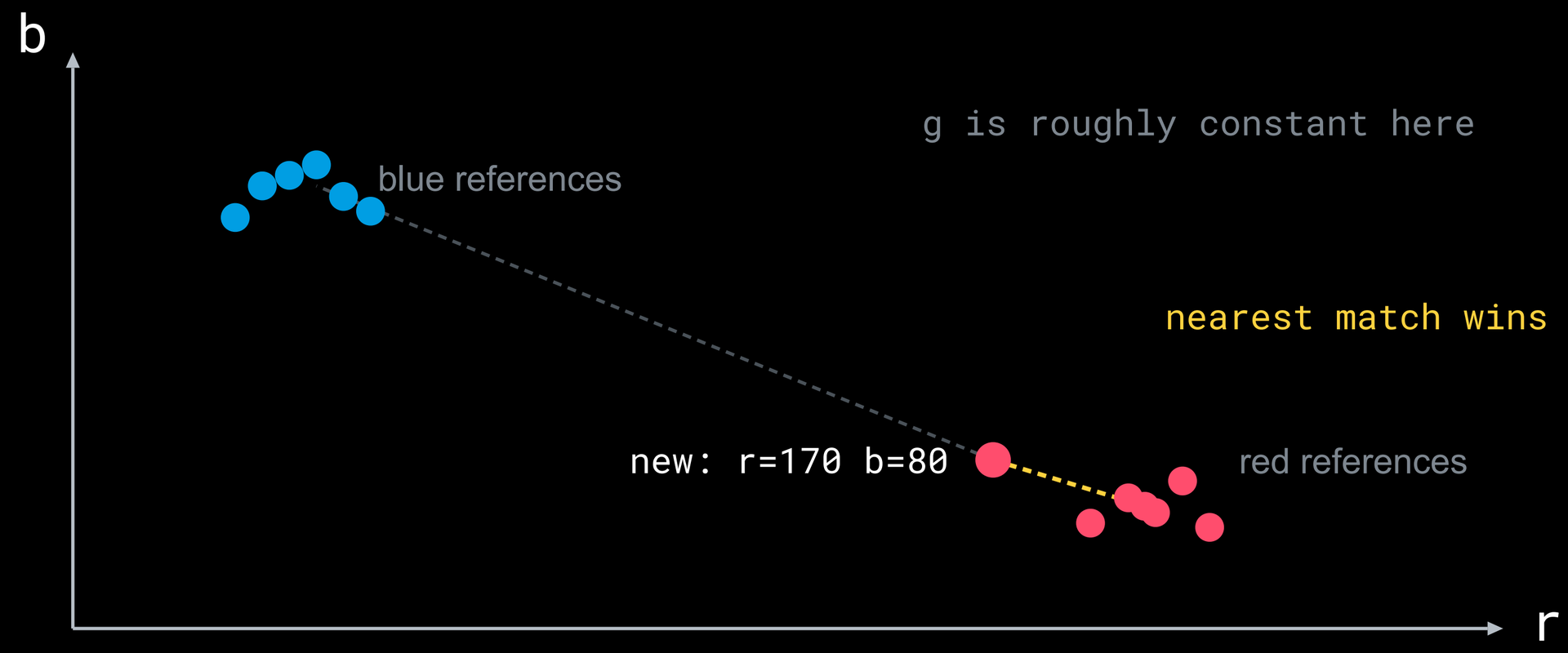
measure known colours many times

calibrate, then classify



measure known colours many times

calibrate, then classify



measure known colours many times

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

```
gain ↑
recognition: 94% → 82%
```

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

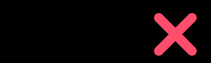
```
gain ↑
recognition: 94% → 82%
```

✓ one variable ✓ enough repetitions ✓ control reading

when the assistant and the data disagree

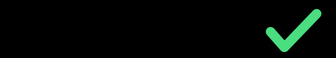
assistant suggestion

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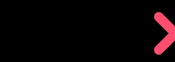
✓ one variable ✓ enough repetitions ✓ control reading

ground truth for your setup

when the assistant and the data disagree

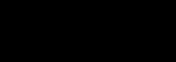
assistant suggestion

"increase the gain —
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your measurement series

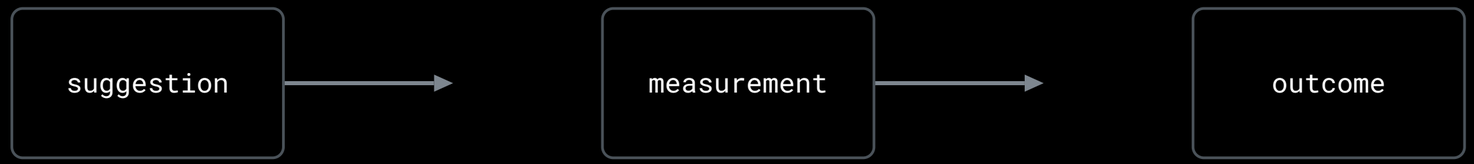
```
gain ↑
recognition: 94% → 82%
```



✓ one variable ✓ enough repetitions ✓ control reading

ground truth for your setup

log of mistakes



if you did not measure it, you do not know it.

one reading is not a measurement series.